# Impact of COVID-19 on ABC Company Parcel Business

## Objectives:

This analysis examines how COVID-19 impacted ABC Company's parcel delivery business.

We aim to:

- Evaluate changes in parcel volumes before and during COVID-19
- Identify when the impact began
- Analyze peak season performance
- Identity how revenue was impacted
- Assess customer growth and behavior across customer segments.

This code cell imports essential Python libraries for data manipulation (`pandas`, `numpy`), visualization (`matplotlib.pyplot`, `seaborn`) and date/time operations (`datetime`). It also configures `seaborn` for better aesthetics and sets the default figure size for `matplotlib` plots. The `%matplotlib inline` command ensures that plots are displayed directly within the notebook output.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.ticker import FuncFormatter
import seaborn as sns
from datetime import datetime

sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 6)
%matplotlib inline
```

This code loads the dataset from the CSV file into a pandas DataFrame.

```
df = pd.read_csv('COVID_Parcel_Business.csv')
```

## 1. Dataset Overview

The dataset contains customer ID, year, week and parcel volume columns. Each row represents weekly shipments allowing trend and comparison analysis.

```
print(f"Shape: {df.shape}")
print(f"Years: {sorted(df['THE_YEAR'].unique())}")
print(f"Unique Customers: {df['FakeCustomerID'].nunique():,}")
df
```

```
Shape: (190719, 4)
Years: [np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021)]
Unique Customers: 1,061
```

| | FakeCustomerID | THE_YEAR | THE_WEEK | VOLUME |
|---|---|---|---|---|
| **0** | 718117 | 2021 | 1 | 8 |
| **1** | 718117 | 2020 | 7 | 257 |
| **2** | 718117 | 2019 | 39 | 141 |
| **3** | 718117 | 2018 | 1 | 14 |
| **4** | 718117 | 2021 | 2 | 192 |
| **...** | ... | ... | ... | ... |
| **190714** | 228893 | 2019 | 13 | 30 |
| **190715** | 228893 | 2019 | 38 | 122 |
| **190716** | 228893 | 2019 | 48 | 99 |
| **190717** | 228893 | 2019 | 20 | 61 |
| **190718** | 228893 | 2019 | 19 | 56 |

190719 rows × 4 columns

## 2.(a) Perfroming Descriptive Statistics on Dataset

This section performs a basic descriptive statistics of our dataset. It gives us the count, mean, standard deviation, min, max and quartile values for FakeCustomerID, THE_YEAR, THE_WEEK and VOLUME columns. This gives a quick overview of the data's distribution.

```
df.describe()
```

| | FakeCustomerID | THE_YEAR | THE_WEEK | VOLUME |
|---|---|---|---|---|
| **count** | 190719.000000 | 190719.000000 | 190719.000000 | 190719.000000 |
| **mean** | 541440.234963 | 2019.395220 | 25.676377 | 1174.911519 |
| **std** | 264861.365663 | 1.068845 | 14.722539 | 11586.730666 |
| **min** | 100771.000000 | 2018.000000 | 1.000000 | 1.000000 |
| **25%** | 308930.000000 | 2018.000000 | 13.000000 | 25.000000 |
| **50%** | 529707.000000 | 2019.000000 | 25.000000 | 58.000000 |
| **75%** | 778800.000000 | 2020.000000 | 38.000000 | 162.000000 |
| **max** | 999362.000000 | 2021.000000 | 53.000000 | 640590.000000 |

## 3. Industry Standard Growth Rate (ISGR)

We are now calculating the Industry Standard Growth Rate (ISGR) based on pre-COVID parcel volumes. This will provide a baseline for normal business growth, allowing us to identify COVID-driven growth later.

The pre-COVID parcel volume increased from **13,122,816 in 2019** to **14,619,017 in 2020**, resulting in an ISGR of **11.4%**.

This indicates that the business was already experiencing steady growth prior to the COVID-19 pandemic.

The ISGR serves as a baseline for normal business growth. Any customer or segment with growth exceeding **11.4%** during the COVID period can be considered to have experienced additional COVID-driven growth.

This distinction ensures that increases in parcel volumes are correctly attributed to the impact of the pandemic rather than normal business expansion.

```
# Pre-COVID periods (Weeks 1-15)
vol_2019_pre = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']<=15)]['VOLUME'].sum()
vol_2020_pre = df[(df['THE_YEAR']==2020) & (df['THE_WEEK']<=15)]['VOLUME'].sum()

# ISGR calculation
ISGR = ((vol_2020_pre - vol_2019_pre) / vol_2019_pre) * 100

print("Pre-COVID Volume 2019 - (W1-W15):", f"{vol_2019_pre:,}")
```

```
print("Pre-COVID Volume 2020 - (W1-W15):", f"{vol_2020_pre:,}")
print("ISGR (%):", round(ISGR, 2))
```

```
Pre-COVID Volume 2019 - (W1-W15): 13,122,816
Pre-COVID Volume 2020 - (W1-W15): 14,619,017
ISGR (%): 11.4
```

## 4. Annual Parcel Volume Analysis and Year-over-Year Change

This section provides the total parcel volume for each year and calculates the Year-over-Year (YoY) percentage change to highlight growth trends in a bit to exploring our dataset.

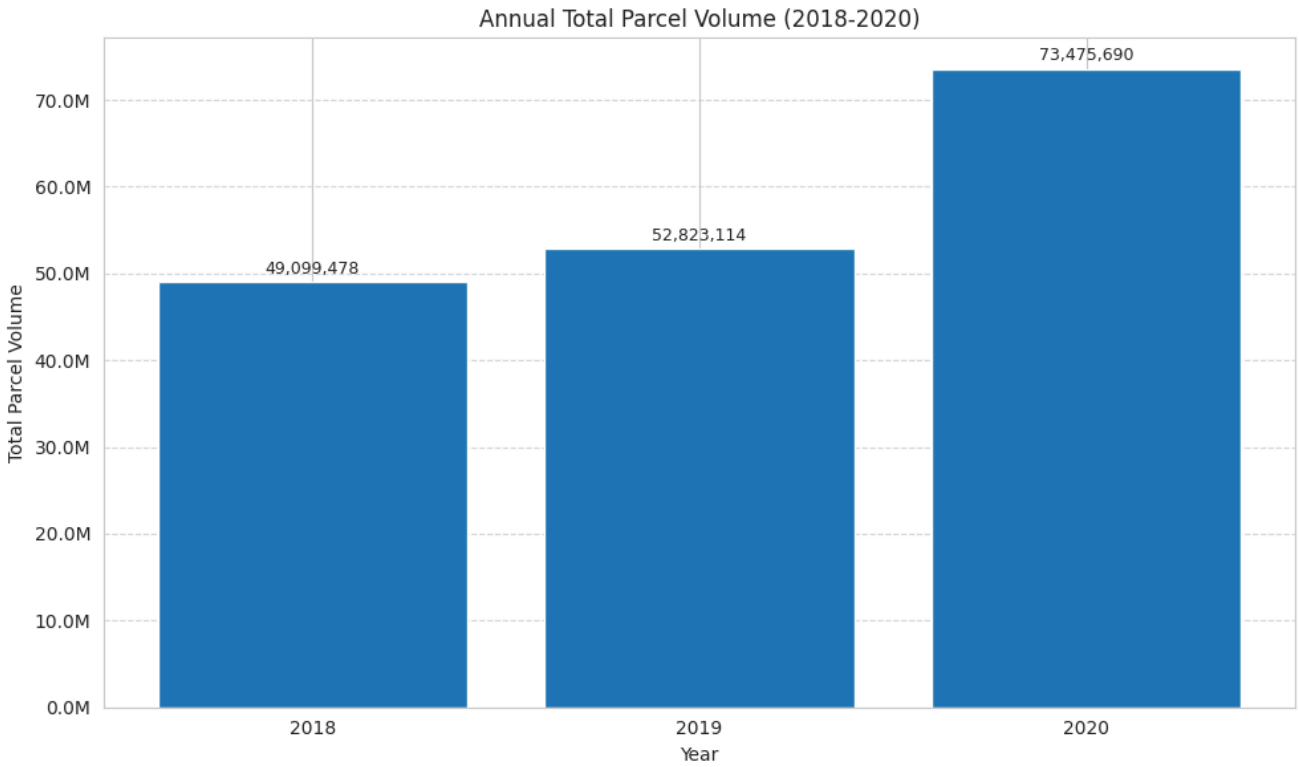
This shows a substantial growth in parcel volume in 2020, likely influenced by the COVID-19 pandemic, followed by a decrease in 2021, possibly indicating a normalization of demand.

Based on the annual parcel volume analysis, it's clear that parcel volumes increased significantly in 2020, indicating a strong growth in demand driven by COVID-19 and the expansion of e-commerce.

```
yearly_volume = df.groupby('THE_YEAR')['VOLUME'].sum().reset_index()
yearly_volume.rename(columns={'VOLUME': 'Total_Volume'}, inplace=True)

# Calculate Year-over-Year Change
yearly_volume['YoY_Change'] = yearly_volume['Total_Volume'].pct_change() * 100

print("Annual Total Volume and YoY Change:")
display(yearly_volume.round(2))

# Filter for years 2018, 2019, and 2020 for the chart
years_to_plot = yearly_volume[yearly_volume['THE_YEAR'].isin([2018, 2019, 2020])]

# Create the bar chart
plt.figure(figsize=(10, 6))
bars = plt.bar(years_to_plot['THE_YEAR'].astype(str), years_to_plot['Total_Volume'])

# Add values on top of the bars
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + (yval * 0.01), f'{yval:,.0f}', ha='center', va='bottom', fontsize=9)

# Add axis titles and chart title
plt.xlabel('Year')
plt.ylabel('Total Parcel Volume')
plt.title('Annual Total Parcel Volume (2018-2020)')
plt.grid(axis='y', linestyle='--', alpha=0.7)

# Format Y-axis to Millions for better readability
def millions_formatter(x, pos):
    return f'{x/1_000_000:.1f}M'

plt.gca().yaxis.set_major_formatter(FuncFormatter(millions_formatter))

plt.tight_layout()
plt.show()
```

```
Annual Total Volume and YoY Change:
```

| | THE_YEAR | Total_Volume | YoY_Change |
|---|---|---|---|
| **0** | 2018 | 49099478 | NaN |
| **1** | 2019 | 52823114 | 7.58 |
| **2** | 2020 | 73475690 | 39.10 |
| **3** | 2021 | 48679668 | -33.75 |




## 5. Customer Segmentation (Customer Groups Information)

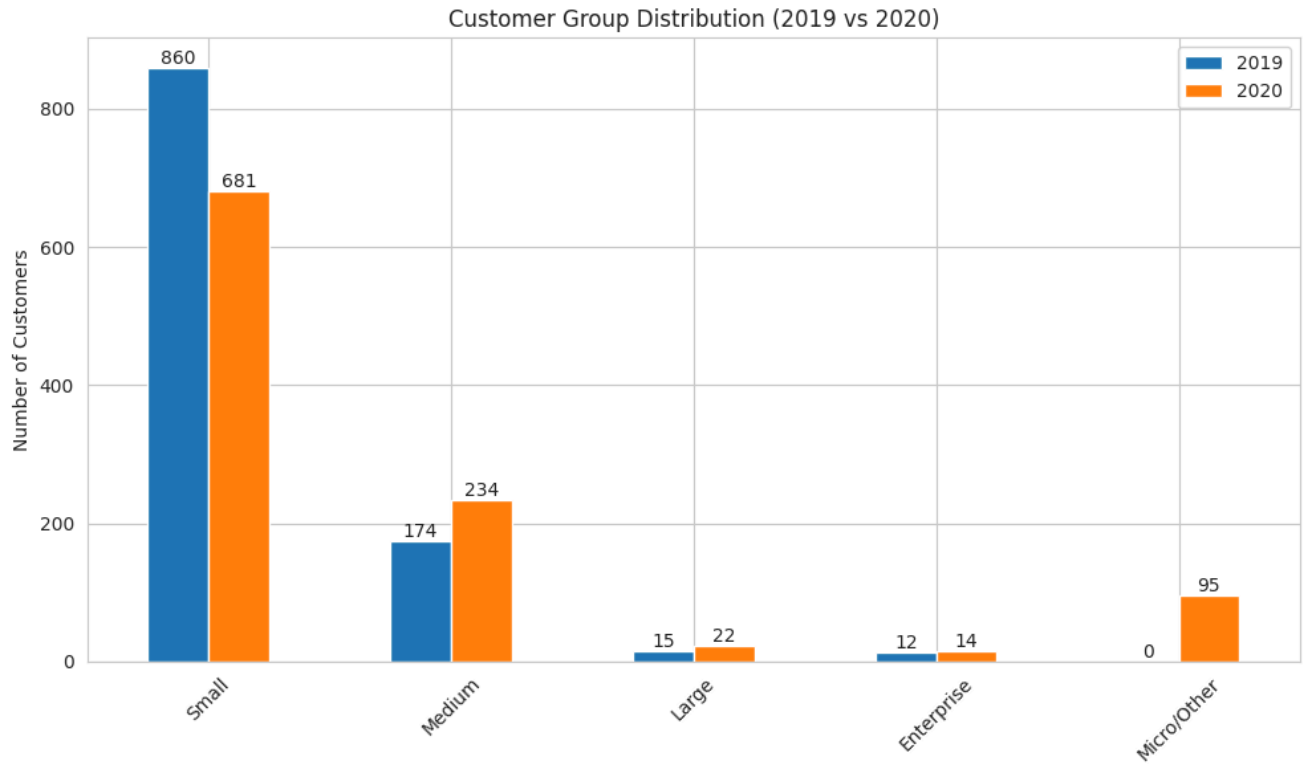
Customers are segmented based on Volume by Customer Segment (2019 vs 2020). This analysis compares parcel volumes across customer segments to assess the impact of COVID-19 in 2019 and 2020.

| Customer Size | Definition | Discount |
|---|---|---|
| Enterprise | 2019 parcel volume >500K pieces | 22% |
| Large | 2019 parcel volume between 200K-500K | 17% |
| Medium | 2019 parcel volume between 10K-200K | 10% |
| Small | 2019 parcel volume between 1K-10K | 4% |

```
# --- Step 1: Aggregate volumes ---
vol_2019 = df[df['THE_YEAR'] == 2019].groupby('FakeCustomerID')['VOLUME'].sum().reset_index()
vol_2019.columns = ['FakeCustomerID', 'Vol2019']

vol_2020 = df[df['THE_YEAR'] == 2020].groupby('FakeCustomerID')['VOLUME'].sum().reset_index()
vol_2020.columns = ['FakeCustomerID', 'Vol2020']

# Step 2: Define grouping logic
def assign_group(v):
    if v > 500_000:
        return 'Enterprise'
    elif v >= 200_000:
        return 'Large'
```

```
    elif v >= 10_000:
        return 'Medium'
    elif v >= 1_000:
        return 'Small'
    else:
        return 'Micro/Other'

# Apply grouping
vol_2019['Group2019'] = vol_2019['Vol2019'].apply(assign_group)
vol_2020['Group2020'] = vol_2020['Vol2020'].apply(assign_group)

#  Step 3: Distribution comparison
dist_2019 = vol_2019['Group2019'].value_counts().rename("2019")
dist_2020 = vol_2020['Group2020'].value_counts().rename("2020")

comparison = pd.concat([dist_2019, dist_2020], axis=1).fillna(0).astype(int)

print("\nCustomer Group Distribution Comparison:")
print(comparison)

#Step 4: Movement / Transition Analysis
movement = pd.merge(vol_2019[['FakeCustomerID', 'Group2019']],
                    vol_2020[['FakeCustomerID', 'Group2020']],
                    on='FakeCustomerID',
                    how='outer')

movement['Group2019'] = movement['Group2019'].fillna('New')
movement['Group2020'] = movement['Group2020'].fillna('Churned')

transition_matrix = pd.crosstab(movement['Group2019'], movement['Group2020'])

print("\nCustomer Movement Matrix:")
print(transition_matrix)

# Plotting with values on top of bars
fig, ax = plt.subplots(figsize=(10, 6)) # Explicitly create figure and axes
comparison.plot(kind='bar', ax=ax) # Plot on the created axes

# Add values on top of the bars
for container in ax.containers:
    ax.bar_label(container, fmt='%d')

ax.set_title("Customer Group Distribution (2019 vs 2020)")
ax.set_ylabel("Number of Customers")
ax.tick_params(axis='x', labelrotation=45) # Use ax.tick_params for label rotation
plt.tight_layout()
plt.show()
```

```
Customer Group Distribution Comparison:
             2019  2020
Small         860   681
Medium        174   234
Large          15    22
Enterprise     12    14
Micro/Other     0    95

Customer Movement Matrix:
Group2020   Churned  Enterprise  Large  Medium  Micro/Other  Small
Group2019
Enterprise        0          12      0       0            0      0
Large             0           2     11       2            0      0
Medium            1           0     11     137            0     25
Small            14           0      0      95           95    656
```




## 6. Pre-COVID vs COVID Comparison

- **Pre-COVID Period: Week 1 to Week 11, 2020**
- **COVID Observation Period: Week 12 to Week 53, 2020**

Please note that a national emergency was declared in Canadan on March 13, 2020 which was on Week 11. The implementation of lockdowns was effected in Week 12 (March 16–22, 2020) by all the Canadian provinces.

Insight: Parcel volumes began experiencing a significant growth during COVID confirming increased delivery demand across Canada.

```
COVID_WEEK_ANALYSIS = 12

# Weekly volumes 2019 vs 2020
weekly = df[df['THE_YEAR'].isin([2019,2020])].groupby(['THE_YEAR','THE_WEEK'])['VOLUME'].sum().reset_index()
w2019 = weekly[weekly['THE_YEAR']==2019].set_index('THE_WEEK')['VOLUME']
w2020 = weekly[weekly['THE_YEAR']==2020].set_index('THE_WEEK')['VOLUME']

# Print weeks around COVID onset
print('Weekly Volume: 2019 vs 2020 (around COVID onset)')
print(f'{"Week":<6} {"2019":>12} {"2020":>12} {"Change":>10}')
print('-'*45)
for w in range(8, 22):
    v19 = w2019.get(w, 0)
    v20 = w2020.get(w, 0)
    chg = (v20-v19)/v19*100 if v19 else 0
```

```
    flag = ' ← COVID onset' if w == COVID_WEEK_ANALYSIS else ''
    print(f'  W{w:<4} {v19:>12,.0f} {v20:>12,.0f} {chg:>+9.1f}%{flag}')
```

```
Weekly Volume: 2019 vs 2020 (around COVID onset)
Week           2019         2020     Change
---------------------------------------------
  W8         764,078      882,633     +15.5%
  W9         803,745      903,362     +12.4%
  W10        792,805      891,398     +12.4%
  W11        879,574      887,335      +0.9%
  W12        864,583      967,436     +11.9% ← COVID onset
  W13        853,762    1,084,829     +27.1%
  W14        817,979    1,346,802     +64.6%
  W15        874,088    1,480,147     +69.3%
  W16        879,169    2,003,866    +127.9%
  W17        912,863    1,914,740    +109.8%
  W18      1,059,240    1,964,582     +85.5%
  W19      1,102,598    2,016,316     +82.9%
  W20        951,212    2,048,931    +115.4%
  W21        874,423    1,842,820    +110.7%
```

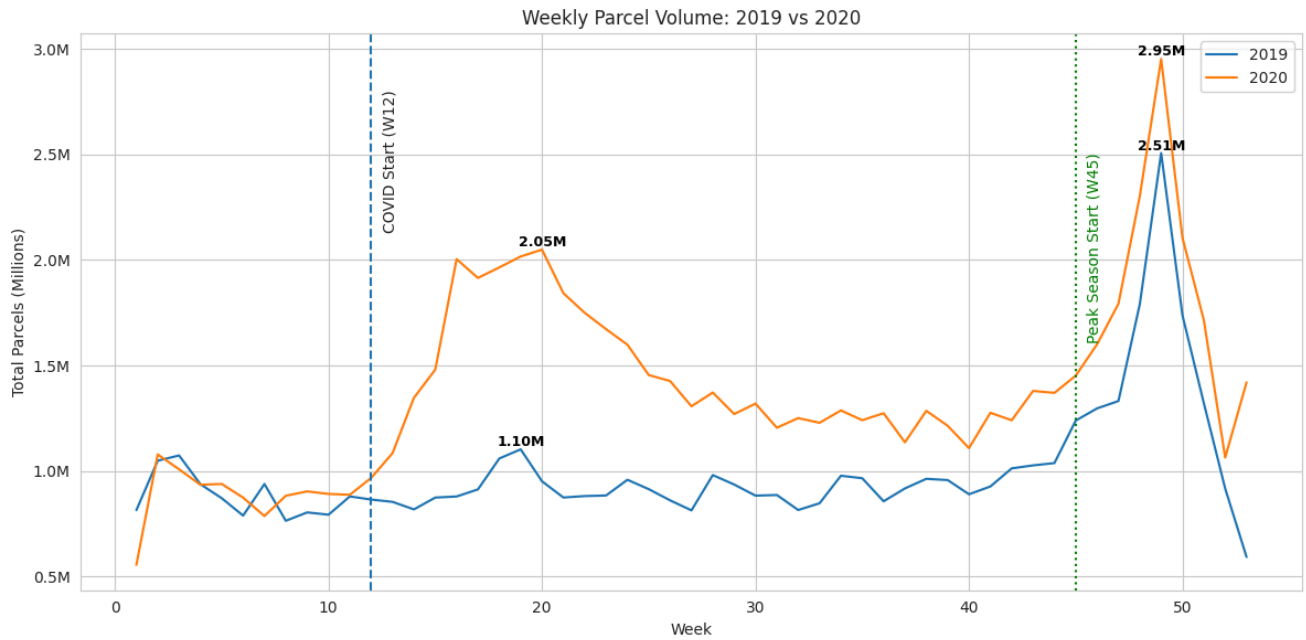
## 6. Weekly Trend Analysis

### Insight:

Volumes begin diverging around Week 12, aligning with COVID lockdowns. This indicates a structural shift in customer behavior.

```
#Filter only 2019 and 2020
df_filtered = df[df['THE_YEAR'].isin([2019, 2020])]

#  Aggregate weekly volume
weekly = df_filtered.groupby(['THE_YEAR', 'THE_WEEK'])['VOLUME'].sum().reset_index()

plt.figure()
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12,8)
%matplotlib inline

# Plot only 2019 and 2020
for year in [2019, 2020]:
    temp = weekly[weekly['THE_YEAR'] == year].sort_values('THE_WEEK')
    plt.plot(temp['THE_WEEK'], temp['VOLUME'], label=str(year))

    # Add values at peaks for specified ranges
    # Range 1: Weeks 10-20
    peak_range1 = temp[(temp['THE_WEEK'] >= 10) & (temp['THE_WEEK'] <= 20)]
    if not peak_range1.empty:
        peak_vol1 = peak_range1['VOLUME'].max()
        peak_week1 = peak_range1[peak_range1['VOLUME'] == peak_vol1]['THE_WEEK'].iloc[0]
        plt.text(peak_week1, peak_vol1, f'{peak_vol1/1_000_000:,.2f}M', ha='center', va='bottom', fontsize=9, color='black', wei

    # Range 2: Weeks 40-50
    peak_range2 = temp[(temp['THE_WEEK'] >= 40) & (temp['THE_WEEK'] <= 50)]
    if not peak_range2.empty:
        peak_vol2 = peak_range2['VOLUME'].max()
        peak_week2 = peak_range2[peak_range2['VOLUME'] == peak_vol2]['THE_WEEK'].iloc[0]
        plt.text(peak_week2, peak_vol2, f'{peak_vol2/1_000_000:,.2f}M', ha='center', va='bottom', fontsize=9, color='black', wei


# Format Y-axis to Millions
def millions(x, pos):
    return f'{x/1_000_000:.1f}M'

plt.gca().yaxis.set_major_formatter(FuncFormatter(millions))

# Get max for positioning
y_max = weekly['VOLUME'].max()

#  COVID marker
plt.axvline(x=12, linestyle='--')
plt.text(12 + 0.5, y_max * 0.95, 'COVID Start (W12)', rotation=90,
         verticalalignment='top')

# Peak Season marker
plt.axvline(x=45, color='green', linestyle=':')
```

```
plt.text(45 + 0.5, y_max * 0.85, 'Peak Season Start (W45)', rotation=90,
         color='green', verticalalignment='top')

#  Labels
plt.title("Weekly Parcel Volume: 2019 vs 2020")
plt.xlabel("Week")
plt.ylabel("Total Parcels (Millions)")
plt.legend()

plt.tight_layout()
plt.show()
```




## 7.Annual Parcel Volume Analysis

### Insight:

Parcel volumes increased significantly in 2020 showing strong demand growth due to COVID-19 and e-commerce expansion compared to 2019.

```
covid_start_week = 11
cust = df.groupby('FakeCustomerID').agg({
    'VOLUME': 'sum'
}).rename(columns={'VOLUME': 'Total_Volume'}).reset_index()

# Volume by key periods
vol_2019 = df[df['THE_YEAR']==2019].groupby('FakeCustomerID')['VOLUME'].sum().rename('Vol_2019')
vol_early2020 = df[(df['THE_YEAR']==2020) & (df['THE_WEEK'] <= 10)].groupby('FakeCustomerID')['VOLUME'].sum().rename('Early2020
vol_covid = df[(df['THE_YEAR']==2020) & (df['THE_WEEK'] >= covid_start_week)].groupby('FakeCustomerID')['VOLUME'].sum().rename(
vol_2020_total = df[df['THE_YEAR']==2020].groupby('FakeCustomerID')['VOLUME'].sum().rename('Vol_2020')

cust = cust.merge(vol_2019, on='FakeCustomerID', how='left').fillna(0)
cust = cust.merge(vol_early2020, on='FakeCustomerID', how='left').fillna(0)
cust = cust.merge(vol_covid, on='FakeCustomerID', how='left').fillna(0)
cust = cust.merge(vol_2020_total, on='FakeCustomerID', how='left').fillna(0)
cust
```

| | FakeCustomerID | Total_Volume | Vol_2019 | Early2020 | Covid_Vol | Vol_2020 |
|---|---|---|---|---|---|---|
| **0** | 100771 | 6252 | 1816 | 347.0 | 1442.0 | 1789.0 |
| **1** | 102180 | 20553 | 5222 | 423.0 | 4394.0 | 4817.0 |
| **2** | 102643 | 8164 | 2670 | 522.0 | 1181.0 | 1703.0 |
| **3** | 102750 | 2747 | 1013 | 184.0 | 221.0 | 405.0 |
| **4** | 103308 | 326088 | 90140 | 16623.0 | 87077.0 | 103700.0 |
| **...** | ... | ... | ... | ... | ... | ... |
| **1056** | 994361 | 7817 | 2608 | 543.0 | 1115.0 | 1658.0 |
| **1057** | 994762 | 2750 | 1073 | 121.0 | 229.0 | 350.0 |
| **1058** | 995173 | 11041 | 1632 | 115.0 | 4805.0 | 4920.0 |
| **1059** | 999141 | 12577 | 3364 | 624.0 | 2995.0 | 3619.0 |
| **1060** | 999362 | 13878 | 3561 | 694.0 | 3637.0 | 4331.0 |

1061 rows × 6 columns

Next steps: Generate code with cust New interactive sheet

## 7. (i) Interpretation of Customer-Level Metrics

The table shows parcel volumes for each customer across pre-COVID and COVID periods.

### Key Insights:

- Many customers exhibit higher parcel volumes during the COVID period compared to earlier periods, indicating increased demand for delivery services.
- For several customers, the majority of their 2020 volume is concentrated during the COVID period, confirming that growth was driven by the pandemic.
- However, not all customers experienced growth—some show lower volumes during COVID, indicating that the impact was uneven across different businesses.
- Larger customers tend to maintain high and stable volumes, while smaller customers show more variability.

Customers in the COVID observation period (W12–53, 2020) are classified as:

- **High Growth**: Growth > ISGR during COVID period
- **Stable**: 0% to ISGR growth during COVID period
- **Declining**: Negative growth
- **Lost**: Had volume in 2019 but zero during COVID period

### Business Interpretation:

Overall, COVID-19 had a strong positive impact on parcel demand, but its effects varied across customers.

This variation highlights the importance of customer segmentation to distinguish between high-growth, stable, and declining businesses during the pandemic.

## 7. (ii) Impact by Customer Group

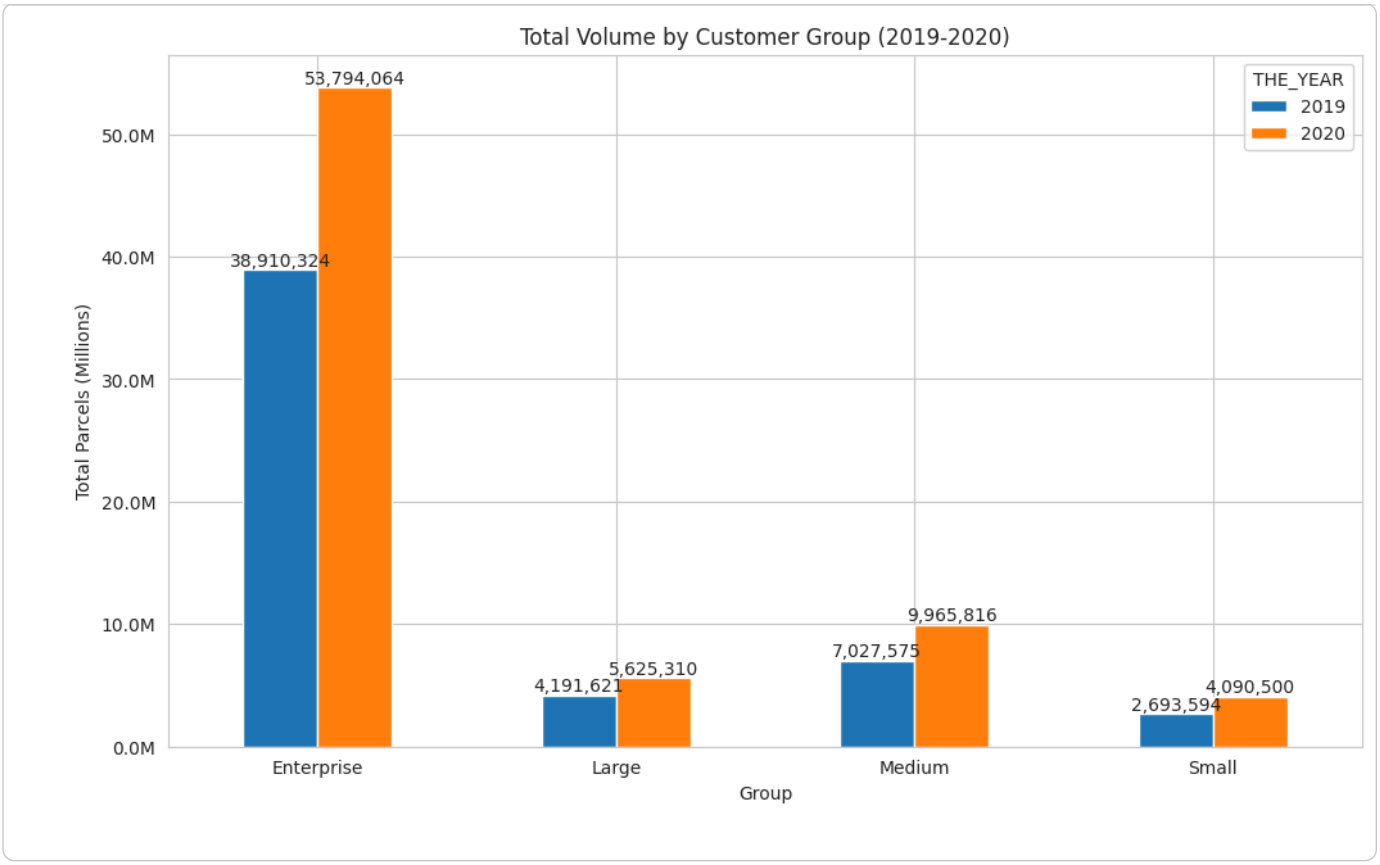
Enterprise customers dominate total volume despite being few in number, while Small customers exhibit the fastest growth rate.

The sharp increase in 2020 across all segments confirms that COVID-19 created a structural shift in demand, rather than a temporary fluctuation.

The decline in 2021 reflects normalization rather than a loss in business strength.

```
# Step 1: Create customer groups from 2019 volume
cust_2019 = df[df['THE_YEAR']==2019] \
    .groupby('FakeCustomerID')['VOLUME'].sum().reset_index()

cust_2019.columns = ['FakeCustomerID','volume_2019']

def group_customer(x):
```

```
    if x > 500000:
        return 'Enterprise'
    elif x > 200000:
        return 'Large'
    elif x > 10000:
        return 'Medium'
    else:
        return 'Small'

cust_2019['Group'] = cust_2019['volume_2019'].apply(group_customer)

# Merge 'Group' into cust
if 'Group' not in cust.columns:
    cust = cust.merge(
        cust_2019[['FakeCustomerID','Group']],
        on='FakeCustomerID',
        how='left'
    )

group_vol = df.merge(cust[['FakeCustomerID', 'Group']], on='FakeCustomerID')

group_year_vol = group_vol[
    group_vol['THE_YEAR'].isin([2019, 2020])
].groupby(['Group', 'THE_YEAR'], observed=False)['VOLUME'].sum().unstack()

# --- Plot ---
ax = group_year_vol.plot(kind='bar', figsize=(12,7))

# Add values on top of the bars
for container in ax.containers:
    ax.bar_label(container, fmt='{:,.0f}')

# --- Format Y-axis to Millions ---
def millions(x, pos):
    return f'{x/1_000_000:.1f}M'

ax.yaxis.set_major_formatter(FuncFormatter(millions))

plt.title('Total Volume by Customer Group (2019-2020)')
plt.ylabel('Total Parcels (Millions)')
plt.xticks(rotation=0)

plt.savefig('volume_by_group_2019_2020.png', dpi=300, bbox_inches='tight')
plt.show()
```




Double-click (or enter) to edit

```
# Ensure threshold is defined
ISGR = 20  # example: 20% growth threshold

COVID_WEEK = 12

# Base: include ALL customers
cust_df = pd.DataFrame(df['FakeCustomerID'].unique(), columns=['FakeCustomerID'])

# Volume calculations
vol_2019 = df[df['THE_YEAR']==2019].groupby('FakeCustomerID')['VOLUME'].sum().rename('Vol2019')
covid_vol = df[(df['THE_YEAR']==2020) & (df['THE_WEEK']>=COVID_WEEK)].groupby('FakeCustomerID')['VOLUME'].sum().rename('CovidVc
same_period_2019 = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']>=COVID_WEEK)].groupby('FakeCustomerID')['VOLUME'].sum().rename('
pre_2019_vol = df[(df['THE_YEAR']==2019) & (df['THE_WEEK']<COVID_WEEK)].groupby('FakeCustomerID')['VOLUME'].sum().rename('PreCc

# Merge all metrics
cust_df = cust_df.merge(vol_2019, on='FakeCustomerID', how='left')
cust_df = cust_df.merge(covid_vol, on='FakeCustomerID', how='left')
cust_df = cust_df.merge(same_period_2019, on='FakeCustomerID', how='left')
cust_df = cust_df.merge(pre_2019_vol, on='FakeCustomerID', how='left')

#  Fill missing values
cust_df[['Vol2019','CovidVol2020','SamePeriod2019','PreCovidVol2019']] = \
    cust_df[['Vol2019','CovidVol2020','SamePeriod2019','PreCovidVol2019']].fillna(0)

#  Classification logic (FIXED)
def classify(row):
    had_vol_before = (row['Vol2019'] > 0) or (row['PreCovidVol2019'] > 0)
    has_covid_vol = row['CovidVol2020'] > 0

    # New customer (no history, appears during COVID)
    if not had_vol_before and has_covid_vol:
        return 'New'

    # Lost customer (had history, disappears during COVID)
    if had_vol_before and not has_covid_vol:
        return 'Lost'

    # Handle zero baseline safely
    if row['SamePeriod2019'] == 0:
```

```
        return 'Stable'  # or 'Undefined' depending on preference

    growth = (row['CovidVol2020'] - row['SamePeriod2019']) / row['SamePeriod2019'] * 100

    if growth > ISGR:
        return 'High Growth'
    elif growth >= 0:
        return 'Stable'
    else:
        return 'Declining'

cust_df['Status'] = cust_df.apply(classify, axis=1)

# OPTIONAL: Add Customer Group (from 2019)
def assign_group(v):
    if v > 500000:
        return 'Enterprise'
    elif v >= 200000:
        return 'Large'
    elif v >= 10000:
        return 'Medium'
    elif v >= 1000:
        return 'Small'
    else:
        return 'Micro/Other'

cust_df['CustomerGroup'] = cust_df['Vol2019'].apply(assign_group)

# Final aggregation
result = cust_df.groupby(['CustomerGroup','Status']).size().unstack(fill_value=0)
pct = result.div(result.sum(axis=1), axis=0) * 100

print('=== Count by Group & Status ===')
print(result)

print('\n=== % Breakdown by Group ===')
print(pct.round(1))
```

```
=== Count by Group & Status ===
Status         Declining  High Growth  Lost  Stable
CustomerGroup
Enterprise             3            7     0       2
Large                  2            9     0       4
Medium                66           78     1      29
Small                299          441    25      95

=== % Breakdown by Group ===
Status         Declining  High Growth  Lost  Stable
CustomerGroup
Enterprise          25.0         58.3   0.0    16.7
Large               13.3         60.0   0.0    26.7
Medium              37.9         44.8   0.6    16.7
Small               34.8         51.3   2.9    11.0
```

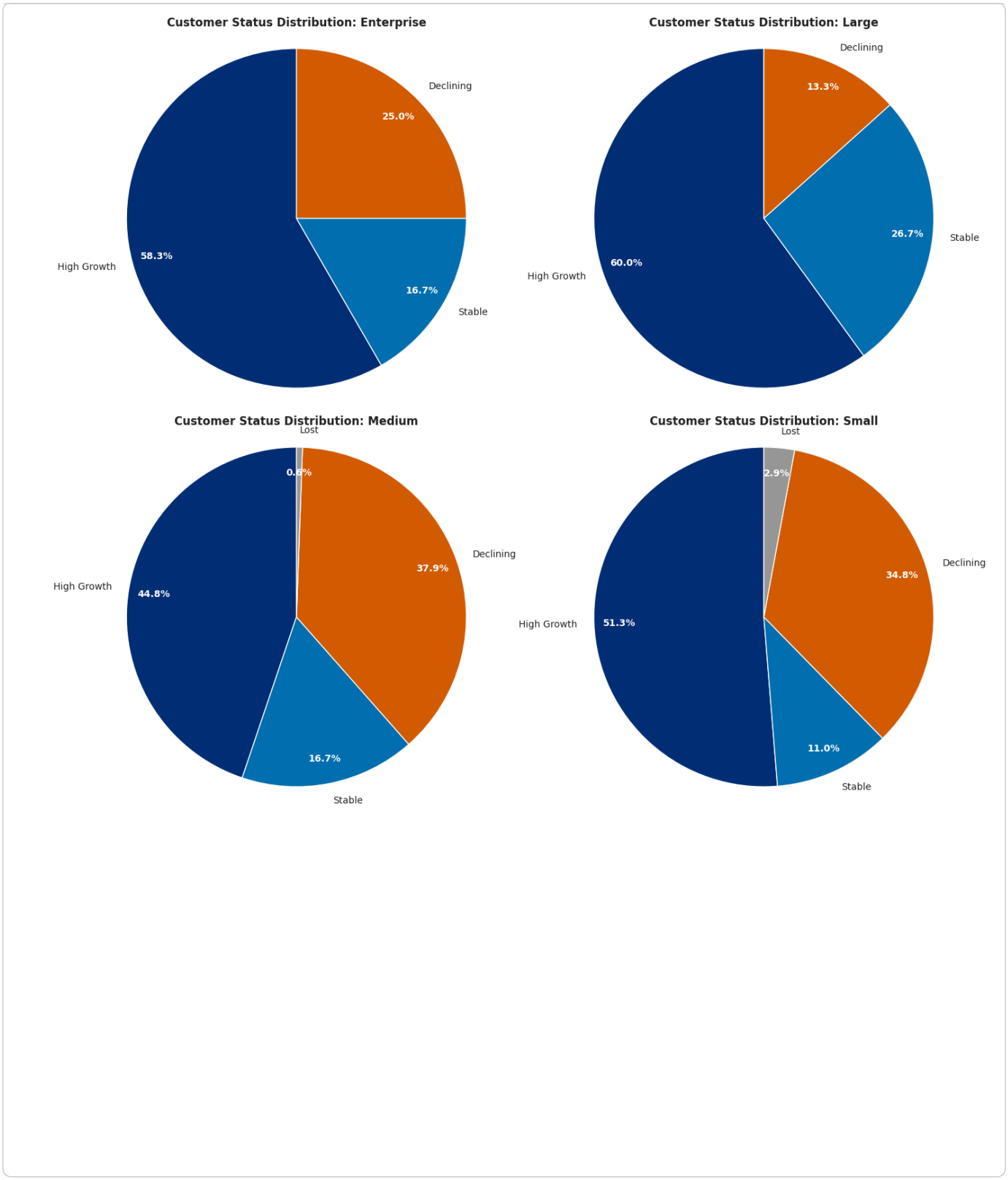
## 7. (iii) Customer Status Chart by Group

This code creates pie charts for each customer segment showing the percentage distribution of customer statuses (High Growth, Stable, Declining, Lost, New).

A consistent colour scheme is used across all charts for easy comparison, with percentages displayed to clearly show how each group was impacted during COVID.

```
colors = {
    'High Growth': '#002E73',   # green
    'Stable': '#0072B2',        # blue
    'Declining': '#D55E00',     # orange-red
    'Lost': '#999999',          # grey
    'New': '#CC79A7'            # purple
}

# Status order for consistent coloring and display
status_order = ['High Growth','Stable','Declining','Lost','New']

# Assuming 'pct' DataFrame is available from the previous cell
# 'pct' contains percentages of customers by CustomerGroup and Status
```

```
# Get unique customer groups from the 'pct' DataFrame index
customer_groups = pct.index.tolist()

# Create a figure and a set of subplots, one for each customer group
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(14, 12))
axes = axes.flatten() # Flatten the 2x2 array of axes for easy iteration

for i, group in enumerate(customer_groups):
    ax = axes[i]

    # Get the status percentages for the current group
    group_percentages = pct.loc[group]

    # Reindex to include all statuses from status_order and ensure consistent ordering
    # and filter out statuses with 0% for cleaner pie chart
    group_percentages = group_percentages.reindex(status_order, fill_value=0)
    group_percentages = group_percentages[group_percentages > 0]

    if not group_percentages.empty:
        pie_colors = [colors.get(s, '#333333') for s in group_percentages.index]

        wedges, texts, autotexts = ax.pie(group_percentages,
                                          labels=group_percentages.index,
                                          autopct='%1.1f%%',
                                          colors=pie_colors,
                                          startangle=90,
                                          pctdistance=0.85,
                                          wedgeprops={'edgecolor': 'white'})

        # Enhance visibility of percentage labels
        for autotext in autotexts:
            autotext.set_color('white')
            autotext.set_fontsize(10)
            autotext.set_fontweight('bold')

        # Ensure labels are clear
        for text in texts:
            text.set_fontsize(10)

    ax.set_title(f'Customer Status Distribution: {group}', fontweight='bold')
    ax.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

# Hide any unused subplots if the number of groups is not exactly 4
if len(customer_groups) < len(axes):
    for j in range(len(customer_groups), len(axes)):
        fig.delaxes(axes[j])

plt.tight_layout()
plt.savefig('chart_customer_status_pies_by_group.png', dpi=150, bbox_inches='tight')
plt.show()
```




## 7. (b) Peak Season Analysis

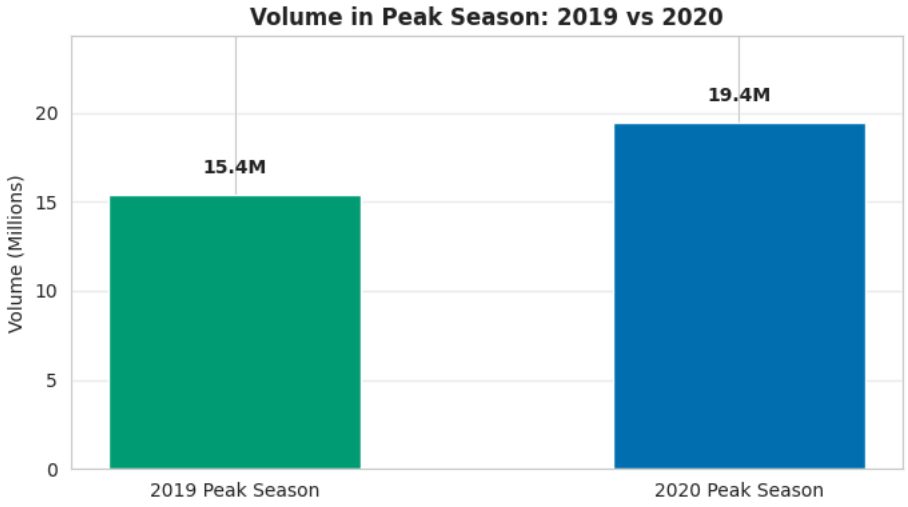
### Insight:

2020 peak season outperformed 2019 driven by increased online shopping.

```
#  Define peak season correctly (cross-year handled clearly)
peak19 = df[
    ((df['THE_YEAR'] == 2019) & (df['THE_WEEK'] >= 45)) |
    ((df['THE_YEAR'] == 2020) & (df['THE_WEEK'] <= 3))
```

```
][‘VOLUME’].sum()

peak20 = df[
    ((df['THE_YEAR'] == 2020) & (df['THE_WEEK'] >= 45)) |
    ((df['THE_YEAR'] == 2021) & (df['THE_WEEK'] <= 3))
][‘VOLUME’].sum()

# Safe % change
change = ((peak20 - peak19) / peak19 * 100) if peak19 != 0 else 0

print(f'Volume in Peak Season 2019: {peak19:,.0f}')
print(f'Volume in Peak Season 2020: {peak20:,.0f}')
print(f'Change: {change:+.1f}%')

#  Prepare values
values = [peak19 / 1e6, peak20 / 1e6]
labels = ['2019 Peak Season', '2020 Peak Season']

#  Plot
fig, ax = plt.subplots(figsize=(7, 4))

bars = ax.bar(
    labels,
    values,
    color=['#009E73', '#0072B2'],  # clean, distinct palette
    width=0.5
)

#  Dynamic scaling
max_val = max(values)
offset = max_val * 0.05

# Bar labels (well positioned)
for bar in bars:
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + offset,
        f'{bar.get_height():.1f}M',
        ha='center',
        va='bottom',
        fontweight='bold'
    )

# Formatting
ax.set_ylabel('Volume (Millions)')
ax.set_title('Volume in Peak Season: 2019 vs 2020', fontweight='bold')
ax.set_ylim(0, max_val * 1.25)
ax.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.savefig('chart_peak_season.png', dpi=150, bbox_inches='tight')
plt.show()
```

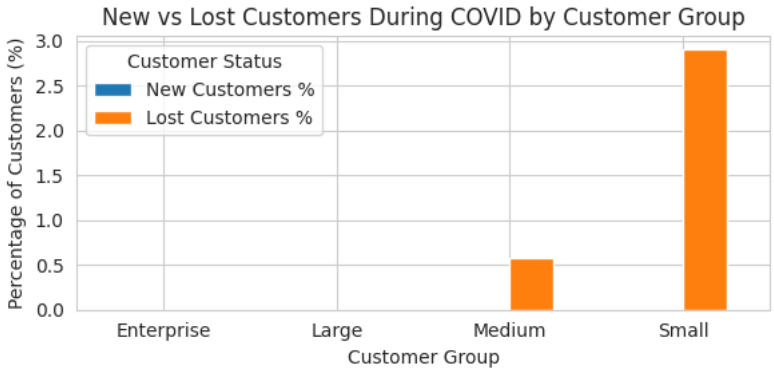
```
Volume in Peak Season 2019: 15,372,691
Volume in Peak Season 2020: 19,442,907
Change: +26.5%
```



## 8. New vs Lost Customers by Group

```
# Define periods
covid_df = df[(df['THE_YEAR'] == 2020) & (df['THE_WEEK'] >= 12)]
df_2019 = df[df['THE_YEAR'] == 2019]

# Customer lists
customers_2019 = set(df_2019['FakeCustomerID'].unique())
customers_covid = set(covid_df['FakeCustomerID'].unique())

# New and lost customers
new_customers = customers_covid - customers_2019
lost_customers = customers_2019 - customers_covid

# Customer group mapping - Corrected to use cust_df
group_map = cust_df.set_index('FakeCustomerID')['CustomerGroup']

# Total customers by group
total_group = group_map.value_counts()

# New customers by group
new_group = group_map.loc[list(new_customers)].value_counts()

# Lost customers by group
lost_group = group_map.loc[list(lost_customers)].value_counts()

# Percent calculation
new_percent = (new_group / total_group * 100).fillna(0)
lost_percent = (lost_group / total_group * 100).fillna(0)

# Combine into one table
new_lost_compare = pd.DataFrame({
    'New Customers %': new_percent,
    'Lost Customers %': lost_percent
}).fillna(0)

print(new_lost_compare)

# Plot comparison chart with reduced figsize
new_lost_compare.plot(kind='bar', figsize=(6, 3))

plt.title('New vs Lost Customers During COVID by Customer Group')
plt.xlabel('Customer Group')
plt.ylabel('Percentage of Customers (%)')
plt.xticks(rotation=0)
plt.legend(title='Customer Status')
plt.tight_layout()
plt.show()
```

```
               New Customers %  Lost Customers %
CustomerGroup
Enterprise                 0.0          0.000000
Large                      0.0          0.000000
Medium                     0.0          0.574713
Small                      0.0          2.906977
```



## 9. Overall impact of COVID on volumes and revenue by customer group

```
# Re-create CustomerGroup in df for consistent analysis within this cell
# Step 1: Create CustomerGroup from 2019 volume
vol_2019_local = df[df['THE_YEAR'] == 2019] \
    .groupby('FakeCustomerID')['VOLUME'].sum().reset_index()

vol_2019_local.columns = ['FakeCustomerID', 'Vol2019']

def assign_group_local(v):
    if v > 500_000:
        return 'Enterprise'
    elif v >= 200_000:
        return 'Large'
    elif v >= 10_000:
        return 'Medium'
    elif v >= 1_000:
        return 'Small'
    else:
        return 'Micro/Other'

vol_2019_local['CustomerGroup'] = vol_2019_local['Vol2019'].apply(assign_group_local)

# Merge CustomerGroup into df (drop if exists to ensure fresh merge)
if 'CustomerGroup' in df.columns:
    df = df.drop(columns=['CustomerGroup'])

df = df.merge(
    vol_2019_local[['FakeCustomerID', 'CustomerGroup']],
    on='FakeCustomerID',
    how='left'
)
df['CustomerGroup'] = df['CustomerGroup'].fillna('New/Unclassified')

price_per_parcel = 22

# Define comparison periods
# Pre-COVID baseline = 2019 full year
pre_covid = df[df['THE_YEAR'] == 2019]

# COVID period = 2020 full year
covid = df[df['THE_YEAR'] == 2020]

# Total volume by customer group
# Now, pre_covid and covid already have the 'CustomerGroup' column due to the above steps
pre_volume = pre_covid.groupby('CustomerGroup')['VOLUME'].sum()
covid_volume = covid.groupby('CustomerGroup')['VOLUME'].sum()

# Combine results
impact_group = pd.DataFrame({
    '2019 Volume': pre_volume,
    '2020 Volume': covid_volume
}).fillna(0)

# Volume change
impact_group['Volume Change'] = impact_group['2020 Volume'] - impact_group['2019 Volume']
impact_group['Volume Change %'] = (
    impact_group['Volume Change'] / impact_group['2019 Volume'] * 100
).replace([float('inf'), -float('inf')], 0).fillna(0)

# Revenue estimate
impact_group['2019 Revenue'] = impact_group['2019 Volume'] * price_per_parcel
impact_group['2020 Revenue'] = impact_group['2020 Volume'] * price_per_parcel
impact_group['Revenue Change'] = impact_group['2020 Revenue'] - impact_group['2019 Revenue']
impact_group['Revenue Change %'] = (
    impact_group['Revenue Change'] / impact_group['2019 Revenue'] * 100
).replace([float('inf'), -float('inf')], 0).fillna(0)

# Display table
impact_group
```

| CustomerGroup | 2019 Volume | 2020 Volume | Volume Change | Volume Change % | 2019 Revenue | 2020 Revenue | Revenue Change | Revenue Change % |
|---|---|---|---|---|---|---|---|---|
| **Enterprise** | 38910324 | 53794064 | 14883740 | 38.251391 | 856027128 | 1183469408 | 327442280 | 38.251391 |
| **Large** | 4191621 | 5625310 | 1433689 | 34.203689 | 92215662 | 123756820 | 31541158 | 34.203689 |
| **Medium** | 7027575 | 9965816 | 2938241 | 41.810169 | 154606650 | 219247952 | 64641302 | 41.810169 |
| **Small** | 2693594 | 4090500 | 1396906 | 51.860303 | 59259068 | 89991000 | 30731932 | 51.860303 |

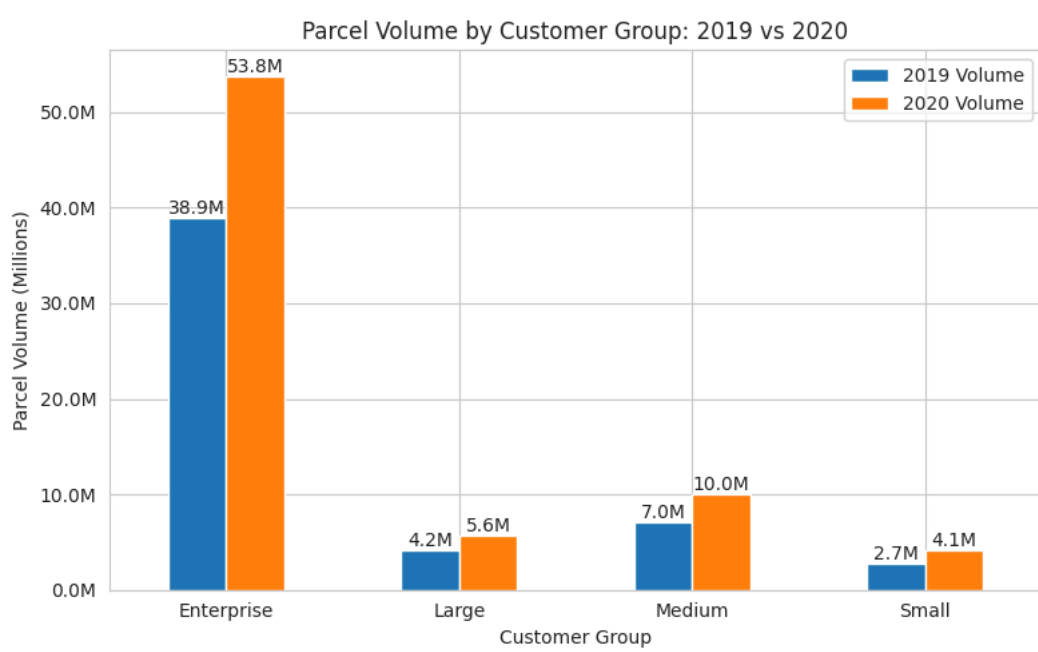
Next steps: Generate code with `impact_group` New interactive sheet

All customer groups grew in 2020, with Small and Medium showing the highest percentage increases, while Enterprise drove the largest overall volume.

Revenue followed the same trend as volume, indicating growth was driven by increased shipping activity across all segments.

volume impact

```
ax = impact_group[['2019 Volume', '2020 Volume']].plot(kind='bar', figsize=(8,5))

# Add values on top of the bars with estimated numbers in millions
for container in ax.containers:
    ax.bar_label(container, fmt=lambda x: f'{x/1_000_000:.1f}M' if x > 0 else '')

plt.title('Parcel Volume by Customer Group: 2019 vs 2020')
plt.xlabel('Customer Group')

# Define formatter for millions (reusing logic from previous cells)
def millions_formatter(x, pos):
    return f'{x/1_000_000:.1f}M'

# Apply formatter to y-axis
ax.yaxis.set_major_formatter(FuncFormatter(millions_formatter))

plt.ylabel('Parcel Volume (Millions)') # Update y-label for clarity
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

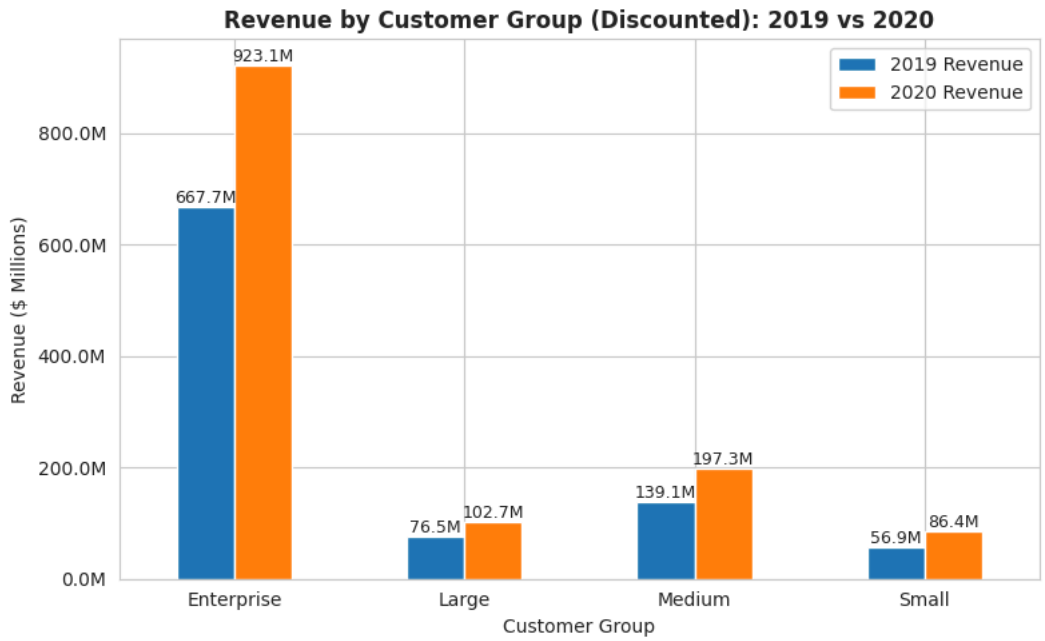



## revenue impact

Revenue increased across all customer groups in 2020, following the same pattern as parcel volume. Enterprise contributed the highest total revenue, while Medium and Small segments showed strong growth. This confirms that revenue growth was driven by increased shipment volume.

```
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import FuncFormatter

# Step 1: Create Customer Groups (based on 2019)
vol_2019 = df[df['THE_YEAR'] == 2019] \
    .groupby('FakeCustomerID')['VOLUME'].sum().reset_index()

vol_2019.columns = ['FakeCustomerID', 'Vol2019']

def assign_group(v):
    if v > 500_000:
        return 'Enterprise'
    elif v >= 200_000:
        return 'Large'
    elif v >= 10_000:
        return 'Medium'
    elif v >= 1_000:
        return 'Small'
    else:
        return 'Micro/Other'

vol_2019['CustomerGroup'] = vol_2019['Vol2019'].apply(assign_group)


# Step 2: Merge safely

if 'CustomerGroup' in df.columns:
    df = df.drop(columns=['CustomerGroup'])

df = df.merge(
    vol_2019[['FakeCustomerID', 'CustomerGroup']],
    on='FakeCustomerID',
    how='left'
)

df['CustomerGroup'] = df['CustomerGroup'].fillna('New/Unclassified')


# Step 3: Apply Pricing (Discounted)

price_map = {
    'Enterprise': 22 * (1 - 0.22),
    'Large': 22 * (1 - 0.17),
    'Medium': 22 * (1 - 0.10),
    'Small': 22 * (1 - 0.04),
    'Micro/Other': 22,
    'New/Unclassified': 22
}

df['price'] = df['CustomerGroup'].map(price_map)
df['revenue'] = df['VOLUME'] * df['price']


# Step 4: Build impact_group table

impact_group = df[df['THE_YEAR'].isin([2019, 2020])] \
    .groupby(['CustomerGroup', 'THE_YEAR'])['revenue'] \
    .sum() \
    .unstack() \
    .fillna(0)

impact_group.columns = ['2019 Revenue', '2020 Revenue']


# Step 5: Plot

fig, ax = plt.subplots(figsize=(8,5))

impact_group[['2019 Revenue', '2020 Revenue']].plot(
    kind='bar',
```

```
    ax=ax
)

# Add labels (in Millions)
for container in ax.containers:
    ax.bar_label(
        container,
        fmt=lambda x: f'{x/1_000_000:.1f}M' if x > 0 else '',
        fontsize=9
    )

# Axis formatting
def millions_formatter(x, pos):
    return f'{x/1_000_000:.1f}M'

ax.yaxis.set_major_formatter(FuncFormatter(millions_formatter))

# Titles
ax.set_title('Revenue by Customer Group (Discounted): 2019 vs 2020', fontweight='bold')
ax.set_xlabel('Customer Group')
ax.set_ylabel('Revenue ($ Millions)')
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```




## Summary of Total Discounted Revenue

**2019 Total Revenue:** $940,274,849.58

**2020 Total Revenue:** $1,309,538,815.64

This analysis shows a substantial increase in total discounted revenue from 2019 to 2020, further reinforcing the significant positive impact of the COVID-19 pandemic on parcel business, even when considering group-specific discounts.

```
# Step 1: Create CustomerGroup from 2019 volume
vol_2019 = df[df['THE_YEAR'] == 2019] \
    .groupby('FakeCustomerID')['VOLUME'].sum().reset_index()

vol_2019.columns = ['FakeCustomerID', 'Vol2019']

def assign_group(v):
    if v > 500_000:
        return 'Enterprise'
    elif v >= 200_000:
        return 'Large'
    elif v >= 10_000:
```

```
        return 'Medium'
    elif v >= 1_000:
        return 'Small'
    else:
        return 'Micro/Other'

vol_2019['CustomerGroup'] = vol_2019['Vol2019'].apply(assign_group)

# Step 2: Merge safely
# Drop if exists to avoid duplication errors
if 'CustomerGroup' in df.columns:
    df = df.drop(columns=['CustomerGroup'])

df = df.merge(
    vol_2019[['FakeCustomerID', 'CustomerGroup']],
    on='FakeCustomerID',
    how='left'
)
```